

Impact of ungulates on forest vegetation and its dependence on the silvicultural system

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Abstract

In order to obtain an understanding of the forest-ungulate compartment in the ecosystem with the aim of better management, the impact of ungulates on forest vegetation, as well as the impact of habitat structure and dynamics on ungulates (density, distribution etc.) and on the forest's predisposition to game damage were investigated. It is shown that browsing and peeling impact depends markedly on silvicultural techniques.

The attractiveness of habitats for game depends not only on food supply, but also, to a high degree, on food-independent habitat factors such as terrain conditions, climate, edge effect, disturbance and competition impact, and thermal and hiding cover availability. Forests with a badly managed ratio of settling stimulus to available food act as 'ecological traps', where the food needed for the over-abundant game ungulates is taken increasingly by the twig browsing and bark peeling of timber species. In general, one can say that a clear-cut system is attractive to deer and chamois. It is easy to hunt in, but it is susceptible to game damage. In particular, clear cutting in narrow strips and reforestation have a high predisposition to game damage. In contrast, selective silviculture results in a more balanced system with less impact by ungulate game on forest vegetation, though hunting might be more difficult. Higher deer densities need not be associated with greater browsing damage; such damage also depends strongly on the growing-stock target and the silvicultural system.

If forestry practices are 'close to nature', an abundance of ungulates may also result in a greater density of forest regeneration and a better mixture of tree species. The potential impact of ungulates on forest regeneration, man-made disturbances of the ungulate-vegetation system, and silvicultural measures to avoid game damage are discussed.

Keywords: Forestry; Ungulates; Game damage; Wildlife ecology

1. Introduction

It is well known that ungulates can markedly change the forest structure by their impact on vegetation and soil. Foresters are particularly conscious of damage to forests by game (e.g. König, 1976; Mayer,

1984; Pollanschütz, 1984; Eiberle and Nigg, 1987; Gill, 1992). Much less known are the 'positive' ecological effects of ungulates on the forest ecosystem. However, ungulate game needs an ecosystem that is 'close to nature' to fulfil this function in a natural way (Reimoser, 1986). Forestry often unconsciously ignores wildlife habitat management effects, thereby provoking game damage (König and Gossow, 1979; Gossow, 1986). (Otto, 1979) summarized the situation well: 'Only an understanding of the

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impacts of the managed forest on game allows, conversely, the impacts of game on the forest to be correctly attributed and, in consequence, potential solutions to conflicts to be sought.'

In this key-note address, some related experimental results about plant–ungulate–silviculture interactions are presented to give incentive for further investigations. In addition, some theoretical aspects of these interactions are discussed. The assessment of game damage as well as of a forest's predisposition to game damage are also considered. Other interactions in the ecosystem, e.g. the carnivore–ungulate compartment, cannot be dealt with here.

2. Materials and methods

Some results of studies being undertaken in Austria on the relationships between ungulates and their environment are presented. These concern in particular roe deer (*Capreolus capreolus*), as well as to some extent red deer (*Cervus elaphus*) and chamois (*Rupicapra rupicapra*). For details about the areas investigated and the methods used (habitat, vegetation and game-damage analyses; track mapping, pellet counts and deer observations for game-distribution analysis), the references mentioned in the text should be consulted, in particular Reimoser (1986); Fischer and Gossow (1987); and Reimoser and Zandl (1993).

Three groups of studies have to be differentiated.

Group 1: Studies of the development of forest regeneration with and without influence of ungulates (referred to in Section 3.1) are being undertaken in the Austrian federal provinces of Vorarlberg and Upper Austria [Reimoser and Suchant, 1992]. Some 2000 defined, 25 m² browsed areas and comparable neighbouring unbrowsed areas (fenced enclosures) have been set up in different forest communities. At 3-yearly intervals, the number, species, height and form of the young trees in each matching pair of areas are ascertained. First results concerning the impacts of ungulate game, i.e. red deer, roe deer and chamois, on tree-species diversity, regeneration abundance and regeneration structure are given. The simultaneous presence of all three game species precludes differentiation of browsing behaviour and damage by species.

Group 2: The studies relating to Sections 3.2 to 3.6 are being undertaken in Styria (Koralpe), where only roe deer are present. The results in Section 3.2 and Section 3.6 present a comparison for three different areas:

REGION A: 144 ha, in a relatively natural state; clearing methods mainly marginal-strip, shelter-wood and modified group-selection cutting; natural regeneration of spruce, fir, sycamore, beech etc.; occasional small clear-cuts; mean size of stands 1.4 ha.

REGION B: 102 ha, non-natural; small-area forest management (mean stand size 0.8 ha); mainly clear-cuts with spruce reforestation; little natural regeneration.

REGION C: 264 ha, non-natural; relatively large-area forest management (mean stand size 2.7 ha); clear-cuts with spruce reforestation; nearly no natural regeneration.

Group 3: The game-damage predisposition rating (Section 3.7) is based on results from several different areas in Austria, the Duchy of Liechtenstein, and Switzerland (e.g. Reimoser, 1986, 1988a, 1996). The results relate to red deer, roe deer, and chamois.

3. Results

Some examples of the effect of silvicultural system on the interaction between forest vegetation, ungulate game and game damage are given below. Details may be found in Reimoser (1986); Fischer and Gossow (1987), and Reimoser and Zandl (1993).

3.1. Ungulate impact on forest regeneration

The impact of ungulate game on tree species in different areas (Section 2, Group 1) have variously been found to result in: (i) a decrease in diversity and/or abundance; (ii) an increase in diversity and/or abundance; (iii) changes in structure without change in diversity or abundance; or (iv) no ascertainable influence. Which of these reactions was realized at a certain place and time basically depended on (i) the type of 'disturbance' (e.g. the nature, intensity and duration of the impact of ungulates on soil and plants), and (ii) the 'reaction' of the respective system (e.g. soil and plants). The type of reaction depended on the initial situation at the time

of the disturbance (soil status, germ conditions, vegetation density, species composition, browsing activity, browsing tolerance, competition between plant species, existing seed trees, light and growth conditions, direction of development, etc.). The initial situation also depended on the type of forest management.

Thus the effect of ungulates — even when they are a constant ‘disturbing factor’ (e.g. same deer density) — can lead to totally different results. For example, an increase of diversity and abundance might turn into a decrease or vice versa. Thus the direction of development of an ecosystem, its self-dynamics and possible feedbacks, can be markedly modified. Human impact (e.g. silvicultural measures, pollution) often stimulates the effect of ungulates on the ecosystem (Reimoser, 1986; Ellenberg, 1988). Game damage is a consequence.

3.2. Habitat structure, roe-deer density and browsing damage — a synopsis

Key game-ecological and hunt-control data are presented (Table 1) for three differently managed forest regions of a game preserve that were comparable in terms of site (*Abieti-Fagetum*, i.e. spruce-fir-beech forest) as well as game management and disturbance factors (Reimoser, 1986).

In comparing the three regions (see Section 2, Group 2), the following needs to be emphasized: (i) the highest game density (mean of 4 years of roe-deer observations, track-density and pellet-group data) was found not in the area with the greatest food supply (Region A), but in the area with the greatest density of edge zones, i.e. the area with the greatest biotope attractiveness (Region B). This was valid for all years and seasons; (ii) the greatest recorded game

Table 1

Comparison of characteristic game-ecological and hunting data ^a for three regions studied that have different silvicultural management systems. For a description of the three regions, A, B and C, see text

Characteristic (components considered / variable measured)		Region A	Region B	Region C
Hiding cover (visibility as affected by vegetation and terrain relief)	(index) ^b	17.26	20.06	8.73
Thermal cover (stand canopy density, tree species, terrain relief)	(index) ^b	17.10	20.77	45.18
Living area (stand type, stand size, terrain relief)	(index) ^b	18.61	41.08	11.92
Edge density (length of stand edges per unit area)	(m ha ⁻¹)	158	221	97
Food supply (percentage of effective grazing area per unit area)	(%)	35.40	27.20	12.30
Percentage browsed, spruce (% of trees)	(%)	0.76	4.22	22.30
Percentage browsed of admixed tree species (% of trees)	(%)	41.9	52.0	57.2
Browsing damage, spruce (percentage browsed, total trees per hectare)	(index) ^c	0.05	0.58	1.66
Browsing damage of admixed tree species (percentage browsed, total trees per hectare)	(index) ^c	78.08	101.52	1181.82
Roe-deer density (deer observed per hour, track density, pellet-group density)	(index) ^d	102	128	70
Roe deer observed per hour (90% still hunt from stand, 10% stalking)	(h ⁻¹)	0.62	0.82	0.50
Roe deer identified (males + females + fawns per all deer observed)	(%)	78.3	86.4	91.4
Sex ratio observed (males: females, without fawns)	(ratio)	0.43	0.57	0.70
Yearling bucks observed (% of all bucks)	(%)	39	35	26
Fawn rate observed in autumn (fawns per female, without yearling does)	(ratio)	1.22	0.76	0.67
Culling per 100 ha (heads of roe deer annually)	(a ⁻¹)	7.6	8.4	3.5
Time-expended in culling per head of roe deer killed (during hunting season, V–XII)	(h)	22.2	15.9	20.8
Intensity of culling (roe deer killed per roe deer observed)	(%)	7.2	8.1	12.4

^a Annual or mean annual data unless otherwise indicated.

^b The higher the value, the better for roe deer (for details, see Reimoser, 1986).

^c This index is derived from $B/(10^{-4} \times N)$, where B is the percentage of top-browsed trees on an annual basis, and N is the total number of young trees per hectare.

^d This index is derived from $\frac{1}{3}(D_n(\%) + T_n(\%) + P_n(\%))$. Here $D_n(\%)$ is the deer observed per hour for Region n ($n = A, B$ or C) as a percentage of the mean of the deer observed per hour in all regions $\left(\frac{1}{3} \sum_{n=A}^C D_n\right)$. $T_n(\%)$ is a similar figure for track density per day and 100 m control line, $P_n(\%)$ for pellet group density per year and 100 m² sampling area.

damage was in the areas with the lowest game density (Region C). Results even indicate a negative correlation between game density and browsing damage (the latter being primarily a function of forest structure).

The key factors influencing the distributions of game and game-damage in a biotope can, in simplified form, be stated to be the available grazing, the food-independent settling stimulus and the relationship between them (see Sections 4.1 and 4.2); the susceptibility of regeneration to browsing damage plays an important role (see below). The settling stimulus is primarily influenced by the edge effect (includes hiding cover), availability of thermal cover, and terrain relief. This appears valid until forage resources become the limiting factor (Reimoser, 1986).

Clear cutting brings the danger of excessive game damage, in particular if clear cuts are smaller than about 2 ha. With this type of forest management, the economic (damage-dependent) carrying capacity of the forest for roe-deer lies much further below its biological (habitat-dependent) carrying capacity than is the case for natural regeneration. Thus an attempt to introduce lasting game-density regulation by culling is markedly handicapped since, in relation to the food supply, the 'over-optimized' settling stimulus causes the game density to revert to economically unacceptable high levels.

The structure and dynamics of roe-deer stock, as well as the ease of hunting, depend markedly on forest management (Table 1). It was noted that the culling intensity (deer killed as a percentage of deer observed) was highest in Region C, the region with the most damage but the lowest game density. Further culling to reduce damage is hardly feasible since, with increasing hunting pressure, the ability to observe deer when so sparsely distributed deteriorates rapidly (Reimoser, 1986), i.e. the time needed to cull becomes excessive. In situations similar to that in Region C, even a few roe deer can inflict severe browsing damage in the — from a game ecological point of view — destabilized forest environment, in particular when they feel threatened by increased hunting pressure.

Forest management that is 'close-to-nature' results in a reduction of the food-independent settling stimulus of a biotope and an increase in the availabil-

ity of various types of food (i.e. grazing etc.). Hence, the predisposition of a young stand to browsing damage is greatly reduced, while the robustness of the forest in respect of game damage is markedly improved. Of particular significance is the better functionality, i.e. the improved exploitation of natural ecological control mechanisms operating between ungulate game and their environment. In turn, the need for time- and cost-intensive interventions (various protective and regulatory measures) required to limit damage is reduced.

3.3. Edge effect

This paper considers only the effect of inner-forest edges and terrain ridges as they induce game to frequent a given forest area (Fig. 1). Track distribution, pellet groups, resting places and game paths were counted and tracks mapped in relation to forest edge zones. With roe deer, the greater the number of optically distinct edges (e.g. at clear cuts), the greater the abundance of deer, even when the edges have only a poor food supply. The ratio between food (vegetation cover underneath the browse line as a forage resource) and the edge situation (settling stimulus) changes within an age-class structured clear-cut forest in a negative way, because the settling attractiveness of edge zones becomes too high as compared with the available forage resources. Therefore food supply becomes the main or only limiting factor (often more important than hunting as a regulatory device), and browsing damage increases. Studies showed that roe-deer prefer a mosaic of interconnected but different small areas with distinct stand edges to provide for their different needs (e.g. orientation, cover, food, shelter), rather than the homogeneous multipurpose biotope resulting from selective silviculture with natural regeneration (Reimoser, 1986; Reimoser, 1994b). Where there are very pronounced edges (e.g. a sharp linear boundary between, for example, a spruce thicket and a beech stand), there is a measurable edge effect for a boundary exceeding 5 m in length. When there is only an indistinct transition between different biotopes, there is a rapid reduction in edge effect as the transition depth increases beyond 10 m.

As opposed to the inner forest stand edges, the outer boundaries of the forest, i.e. those bordering on

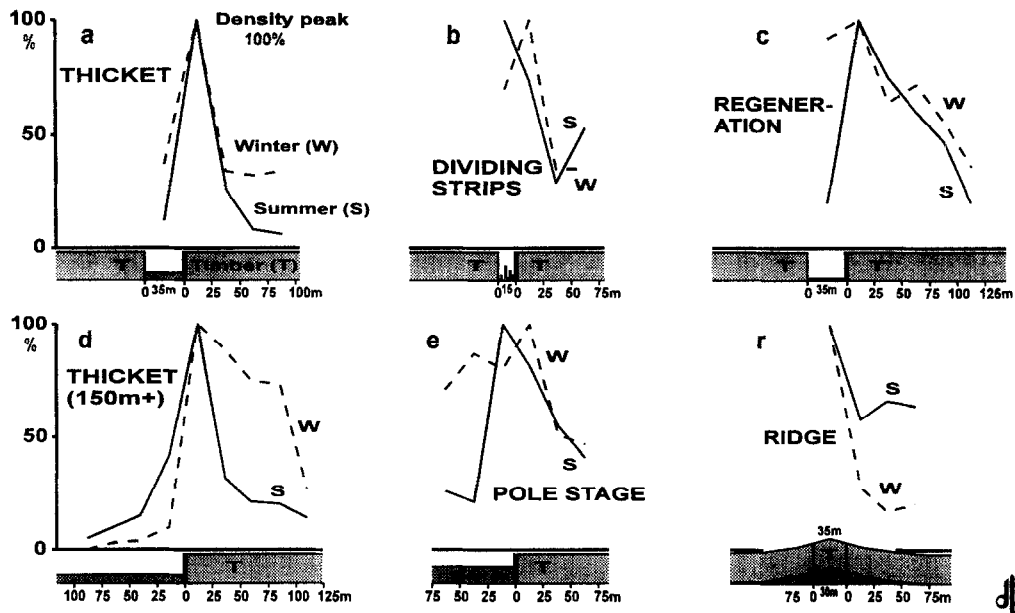


Fig. 1. Effect of stand edges for various stand interfaces (a-e, r) on roe-deer track density. For strip (a-c) and ridge (r) edges, results are given as the mean of data obtained on both sides. Curves are shown for one edge only, the curve for the opposite edge being a mirror image. Track density data are given as a percentage of the maximum value found (for details, see Reimoser, 1994a). Dividing strips are clear cuts 5–15 m wide, with or without natural regeneration, used to partition a forest (interface Type b, between timber-wood stands). Thicket and regenerations strips (Types a and c) are 35 m wide.

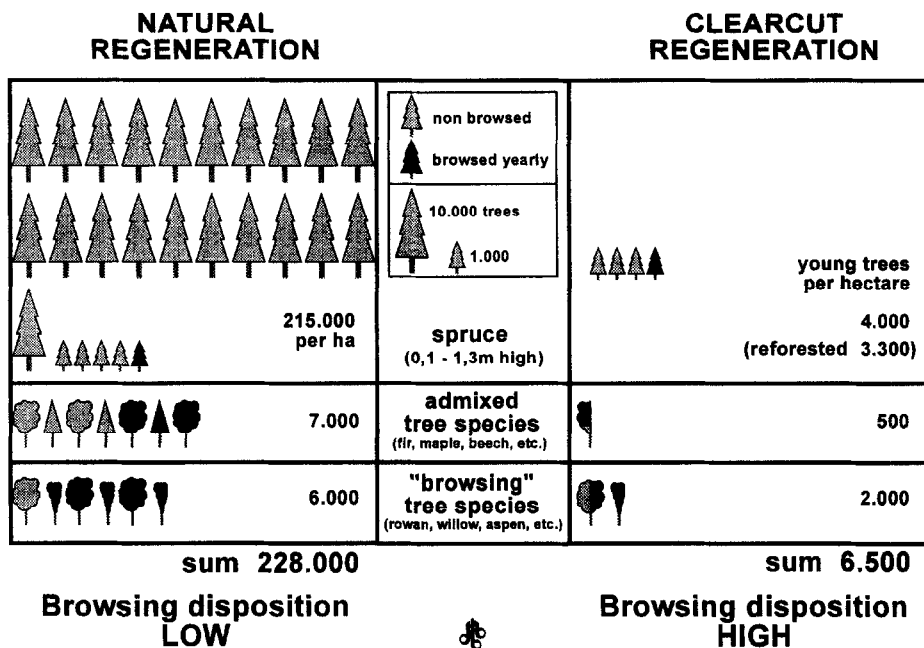


Fig. 2. Influence of the silvicultural system on predisposition to browsing damage of forest regeneration for equal site conditions (Reimoser, 1986). Compared are, on the one hand, an irregular shelterwood system with natural regeneration (surplus trees, many young trees may be browsed without damage to forest, predisposition to damage low) and, on the other hand, a clearcut regeneration system with spruce reforestation (few young trees, browsing predisposition high).

fields and meadows, are less critical in this regard since these are mostly a permanent source of a good and plentiful supply of food.

Edge zones are also attractive for chamois and red deer, whose importance, however, in terms of game distribution is less than that of roe deer. Space precludes more detailed comment. However it has been shown (Reimoser, 1988a) that red deer can be found in the central zones of large-area thickets or pole stands more often than is the case for roe deer. Furthermore, chamois are in particular attracted to open areas (e.g. clear cuts, steep embankments bordering forest roads).

3.4. Density and area of regeneration

The density and composition of forest regeneration affects a forest's predisposition to game damage (Reimoser, 1986; Reimoser, 1992). An example of the game-ecological difference between two silvicultural systems for equal site conditions (clear felling and reforestation, and shelterwood cutting with natural regeneration) is given in Fig. 2 and Fig. 3. The naturally regenerating system has a surplus of young trees per hectare (Fig. 2), so browsing does not result in immediate damage (only 3000 to 4000 intact trees are needed at the thicket stage). In addition, the feeding area for the complete forest working cycle is approximately doubled (Fig. 3).

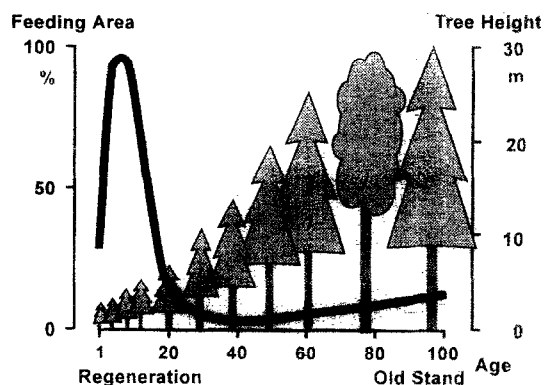
3.5. Browsing percentage as a function of growing space

The trees surrounding the young trees at the centre of dense regeneration groups provide good natural protection against browsing. Solitary trees are browsed much more (Table 2, Fig. 4) (Reimoser, 1986). Trees planted after clearcuts are all 'solitary'; therefore predisposition to browsing is higher than with natural regeneration.

3.6. Food supply, summer and winter

The seasonal availability of food plays a crucial role in respect of a forest's predisposition to game

Artificial Spruce Regeneration after CLEARCUT



NATURAL REGENERATION with Shelterwood

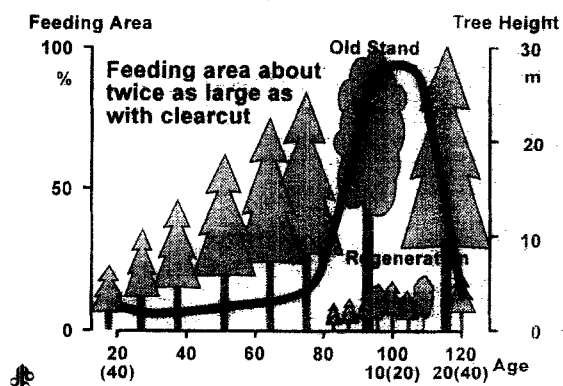


Fig. 3. Influence of the silvicultural system on feeding area during the whole forest working cycle for equal site conditions. The same systems are compared as in Fig. 2. The curve shows the percentage of feeding area available to deer at each stage of forest development (Reimoser, 1986).

Table 2

Percentage of trees browsed (top twig) as a function of surroundings of individual trees (1: solitary tree; 2: trees on one-side only; 3: trees on several sides; 4: tree fully surrounded by other young trees at least up to a distance of 50 cm away)

Tree species	Type of growing space, i.e. surroundings of individual tree			
	1	2	3	4
Fir	44	40	34	13
Sycamore	79	53	53	5
Rowan	87	48	47	9

Table 3

Normalized grazing area (quantity and quality) as a percentage of the area of the Region under summer and winter conditions, as well as differences and winter-to-summer ratios

Region	Summer (%)	Winter (%)	Difference (summer – winter)	Ratio (winter/summer)
A	47.8	23.0	24.8	0.48
B	37.0	17.4	19.6	0.47
C	21.9	2.6	19.3	0.12

damage. If the difference between food availability in summer and in winter is unnaturally high, while there may be good conditions for an increase in game in the summer (drop time, rearing of young, rut), the shortage of food in winter will lead to an increase in game damage. With limited available grazing, a large summer-to-winter difference has a particularly negative effect (Table 3, grazing quotient, Region C, Reimoser, 1986). The food supply available to game in areas with natural regeneration under shelter wood is spatially and temporally better distributed than with clear-cutting.

3.7. A game-damage-predisposition rating for different silvicultural systems

A summary of ratings accorded to various silvicultural practices on montane sites with spruce-fir-beech forests is presented in Table 4 and derives from investigations undertaken in Austria. The rating accords a figure (1 lowest/best; 5 highest/worst) to

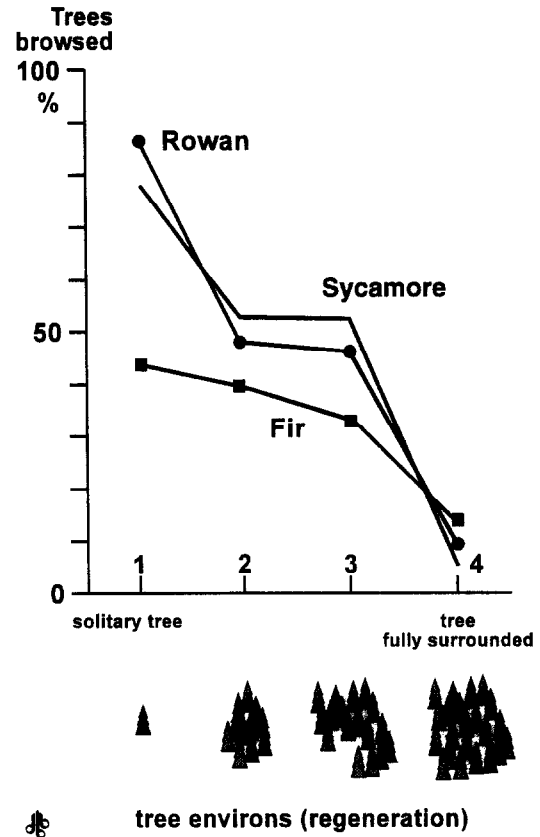


Fig. 4. Percentage of trees browsed (trees up to 1.3 m height with top twig browsed, growing in shelterwood) for rowan (*Sorbus aucuparia*), sycamore (*Acer pseudoplatanus*), and fir (*Abies alba*) as a function of growing space (1: solitary tree; 2: trees on one-side only; 3: trees on several sides; 4: tree fully surrounded by other young trees at least up to a distance of 50 cm away) (See Table 2).

Table 4

Rating of predisposition to browsing and peeling damage of different silvicultural systems for a complete forestry working cycle

Silvicultural system	Predisposition to damage ^a	
	Browsing	Peeling
Small clearcuts, reafforestation	5	5
Large clearcuts (> 2 ha), reafforestation	2 (→ 3)	5
Shelterwood felling, natural regeneration	1	4 (→ 3)
Shelter-strip-group felling, natural regeneration	1	3 (→ 2)
Group selection felling, natural regeneration	2	3 (→ 2)
Single tree selection, natural regeneration	4 (→ 5)	2

^a Index 1: lowest/best; Index 5: highest/worst. Number in parentheses indicates tendency of index.

a forest's predisposition to game damage in respect of browsing and peeling.

4. Discussion

4.1. Some aspects of game damage theory

Game damage by twig browsing and bark peeling is described as being an increasing problem (e.g. Mayer and Ott, 1991; Donaubauer, 1994). However, foresters have difficulties in assessing objectively the damage done by game.

Why is this? It seems that the problem relates to an absence of a target for growing-stock. 'Damage' (in the sense of 'a problem caused by an unwanted condition') is an anthropocentric concept used in relation to one or more specific species. To ascertain 'damage' requires that a concrete aim — a desired condition — be defined and compared with the current condition in order to determine whether 'damage' or merely an impact or disturbance by game has in fact occurred. This requires in turn that operational limits and critical loads are specified. From a species-neutral perspective, the term damage has no meaning. A problem needs an owner — it is the problem owner who defines 'damage'. For example, while heavy browsing will be a problem both for the forester and for the trees, other plants, as well as animals and people that seek open spaces, will welcome the demise of the trees.

When does a disturbance of vegetation by ungulates become damage? *Not every twig browsed represents damage to a tree; not every tree damaged represents damage to a stand.*

To ascertain objectively the existence of game damage, the following must be specified: (i) *Is there a need for stand regeneration?* For example, after a polestand has regenerated for an appropriate period after tending, it has no further need for regeneration. Browsing cannot then be counted by the forester as 'damage', because any regeneration would, even without browsing, die off as the shelter of the stand grows dense again. In other words, the browsing impact is now included within 'compensatory mortality' and therefore does not affect the further development of the stand. Some decades later, when the stand needs to regenerate again, the same browsing

impact may truly be 'damage'. The concept of compensatory mortality is usually discussed in connection with animal populations, but it is at least as important for plant populations. Because of this effect, when assessing game damage, the mortality of trees is less important than the number that survive.

(ii) *The operational regeneration target:* A target set at the planning stage might, for example, be 3000 young trees per hectare undamaged with a distribution of at least 20% spruce, 10% fir, 20% beech, the remainder being of the same or other tree species. If, after disturbance by game, there are still enough undamaged trees to fulfil the target, the browsing will not have reached the level of 'damage'. To reiterate, damage to trees does not automatically qualify as damage to a stand. Thus, when calculating game damage, one has at first to focus on the number of undamaged trees. (iii) *The response of trees to browsing:* The effect of browsing depends on the tree species, site conditions, time and intensity of browsing, etc. Again, not every twig browsed qualifies as damage to the tree. (iv) *Determination of the damaging species:* Is the impact truly being inflicted by game animals or by other species (e.g. cattle, sheep, mice, humans)? Only game animals can effect 'game damage', but there are many causes of damage to trees, often with very similar symptoms.

4.2. A forest's predisposition to game damage

During the last 15 years, forest-ungulate interactions have been investigated in several areas of Austria. An attempt was made to determine the main factors causing game damage and the most important silvicultural factors influencing a forest's predisposition to game damage (Reimoser, 1988a; Reimoser, 1992). Viewed from the aspect presented above, that damage to trees does not automatically qualify as damage to a stand, some principal findings are briefly summarized below.

Game damage can be expressed as a function of a target, a pressure of game on plants, and a predisposition of the forest to game damage. If the target growing-stock (or the related regeneration aim) for a forest is uniform, game damage results both from the browsing–peeling–fraying–trampling pressure of game (frequency of plants and trees damaged by

game) and from predisposition of the forest habitat to game damage. Game damage, GD, in forest stands increases with increasing frequency, F , of trees damaged by game as well as by increasing the predisposition, P , of the forest to game damage:

$$GD = F \cdot P \quad (1)$$

F in turn depends on the mean frequency of trees damaged per animal and day, F_m , game density, D , and the staying period, t , in days:

$$F = F_m \cdot D \cdot t \quad (2)$$

The predisposition of forest to game damage is a result of both settling attractiveness (settling stimulus) of the habitat for game and available feeding. P increases with increasing habitat attraction, HA, and decreasing available food, AF; the ratio in the equation:

$$P = \frac{HA}{AF} \quad (3)$$

is crucial. Seasonal food composition and the regeneration density (cf. Sections 3.4–3.6) are important factors influencing AF.

The habitat attraction (HA) for ungulates, on which game density and distribution as well as browsing intensity primarily depend, is determined by both food-independent settling stimuli and food supply. The former were found to play an unexpectedly important role (Reimoser, 1986). These food-independent factors include terrain conditions, climate, edge effect (at least for roe deer, highly food-independent), game disturbance and competition impacts (intra- and interspecific, including man), as well as thermal and hiding cover availability.

The impact of forestry practices on food supply and food-independent factors affects habitat quality, the density, structure and dynamics of ungulate stocks, the predisposition of forest to game damage, and the scale of game damage that arises.

Indeed, forests with a badly managed ratio of settling stimulus to available food (HA/AF), e.g. when silviculture is not 'close to nature', act as 'ecological traps'. In such traps, the food needed for the overabundant ungulate game is taken increasingly, though unnaturally, by twig browsing and bark

peeling of timber species, and game damage is provoked (cf. Sections 3.2, 3.3). It is shown that game density is only one factor that affects game damage; therefore game damage can arise at any game density.

4.3. Some particular aspects of game damage in relation to habitat and species

In forests that are broken into many small areas incorporating meadows, arable fields and/or settlements, the various extraneous influences, e.g. agriculture, interfere with and weaken the habitat-determining effects of forestry practices. In extensive areas of unbroken forest, those practices exert the principal influence on the quality of a habitat for game and on the game's impact on the forest.

Game-species differences: Roe deer, as selective browsers, tend to be solitary animals, especially within a forest. Thus, for these animals, habitat assessment in terms of biotopes and forest structures gives useful results (e.g. Reimoser, 1986). However, red deer and chamois are longer-lived and more social, and they react more sensitively and adaptively to disturbances, habitat changes and harassment, as well as to changes in resources. The effects of hunting and tourism (as reinforcers of long-term hunting pressure) make a simple, solely vegetation-based evaluation of habitat suitability and carrying capacity less efficient (Fischer and Gossow, 1987). This also applies to supplementary feeding.

Winter feeding: Supplementary feeding of red deer results in their markedly changing their avoidance and habitat-use strategies (less true for roe deer). Such feeding as an overwintering strategy is an important steering instrument in alpine red-deer management, but it is not equally efficient everywhere and at all times (e.g. Gossow, 1986; Gossow, 1987; Schmidt, 1992; Schmidt, 1993; Schmidt and Gossow, 1991). Judged by the effectiveness in preventing browsing and bark-peeling, poorly situated feeding sites have often been found to be counterproductive (e.g. Gossow, 1985; Gossow et al., 1987; Sackl, 1992).

Hunting pressure and tourism: The long hunting seasons, reinforced by increasing tourism, result in

corresponding avoidance strategies (e.g. more pronounced night activity; remaining in less accessible thickets, pole stands, on steep slopes etc. during the day). The resulting game damage is fairly well distributed throughout reafforestation areas, caused primarily by roe deer and chamois; thickets, pole and young tree stands are in particular affected by the bark stripping of red deer.

Chamois, which prefer areas at or above the timberline, also react to increased hunting pressure and to the increasing and reinforcing tourism impacts. Their avoidance strategies have caused them to move into forests (the so-called forest chamois), preferring those with cleared areas, e.g. with forest roads, clear-cuts, clearings for ski-runs, and areas deforested by storm damage (e.g. Hamr, 1988; Gossow, 1992). Thus chamois, too, are increasingly contributing to game damage.

Aspects to be considered when forestry practices change to be 'closer to nature': In Austria there is an increasing public demand that forests be rebuilt into more 'natural' ecosystems. This is matched by an obvious readiness to do so on the part of many foresters. In connection with the reduction, in particular, of game damage, such 'rebuilding' requires, in our view, carefully planned implementation.

A forest plantation often starts with reforestation of a clearcut or an afforestation of an abandoned meadow. Thus there is no pioneer phase of a typical initial succession of brush and non-timber trees, herbs and perennials. We note that this type of vegetation offers a very suitable food base for ungulates and serves as a helpful buffer against deleterious browsing of timber-tree saplings growing in the wake of this succession. In this connection, elder stands should be supported as a source of food for game herbivores, especially by upgrading the understorey and ground vegetation layers through improved and better directed thinning and clearing operations (cf. Loidl and Reimoser, 1980; Reininger, 1987; Reimoser, 1988a).

4.4. Promotion of a flexible, integrated pattern of wildlife and forest management

Studies have confirmed that forest management practices have a profound influence on the settling

and feeding of deer. Thus, to limit game damage, the forest manager will have to recognize that forest management is at the same time habitat management, and the interplay of these must be taken into account more consciously and more actively if a better balance between game and forest is to be achieved.

In Central Europe, reduction of ungulate game damage has, up to the present time, usually been the most important driving force for taking wildlife ecology into account as part of forest management. A more game-oriented silvicultural system is needed. Forestry measures would include fewer clearcuts, and adoption of new forest use and regeneration strategies.

However, forestry measures alone cannot solve the problem; they must be combined with measures that coordinate the aims and targets of all stakeholders — foresters, hunters, farmers, and the tourist authorities — with plans coordinated over large enough regions to be relevant for the game species of interest. Game (especially red deer and chamois) do not remain within one preserve (e.g. seasonal migrations) and can, therefore, only be managed expediently by planning control over sufficiently large areas. Such a concept for large area planning with integral management of deer and chamois as well as their habitats has been developed and realized in two Austrian provinces, Vorarlberg and Salzburg (Reimoser, 1988b; Reimoser, 1996).

Some aspects of forestry control measures are discussed briefly below. A more detailed discussion of these and other measures may be found in Crawford (1984); Gossow (1986); Reimoser (1986); Reimoser (1988a).

Choice of silvicultural system: Crawford (1984) asked the question: 'Is good forestry good wildlife management?'

Investigations as well as practical experience have shown that silviculture that is 'close to nature' and properly planned often results in three distinct benefits: (i) good, i.e. more profitable, forestry; (ii) good resistance to game damage; and (iii) a good wildlife habitat with greater wildlife species diversity (e.g. Eiberle, 1979; Loidl and Reimoser, 1980; Crawford, 1984; Reininger, 1987). While many foresters are still of the (wrong!) opinion that clear-cutting is always the more profitable form of management, a steadily growing number are beginning to recognize

its disadvantages. Indeed, it appears that what may be lacking is access to valid information and further training to improve the standards of knowledge and qualifications of foresters and hunters alike.

Wildlife damage prophylactics should be considered and integrated into forest operations, planning and practice more consequently. Unfortunately, Central European foresters have long been, and often still are, of the opinion that these forest-game problems are primarily a matter of reduction of game density. Yet often, where hunting efforts and culling have been intensified, the reduction of game damage has not matched expectations (Gossow, 1986).

Choice of tree species (mixing ratio): It is important to achieve a proper balance between deciduous and winter-green tree species. If in a forest there are only winter-green species such as spruce and fir, having only a poor food supply in the dark stands, though there is excellent thermal and hiding cover, the ratio of settling stimulus to available food, HA/AF, is out of balance. Hence, the predisposition of the forest to game damage, P , is increased, i.e. game damage is provoked. In Central Europe, large areas originally dominated by deciduous trees were afforested only with spruce; the change to include a proper proportion of deciduous species will take several decades. The problems regarding re-establishment of indigenous tree species are well known. For example, without protection, young trees are extraordinarily susceptible to selective browsing.

Forest tending: Thinning of thickets and pole stands allows more light to reach the ground. The food supply increases, while thermal and hiding cover are reduced. Forest tending brings about a more balanced ratio between available food supply and food-independent settling stimulus, and hence the disposition to game damage is reduced.

Stand edges: An important conclusion is that optically striking stand edges should, where possible, be avoided to reduce game damage. But what is the cause of the edge effect? Already in the 1930s, Leopold (1933) had referred to an edge effect and pointed to the increased presence of certain game species at boundaries between different biotopes. Where an explanation could be ventured, he believed it was linked to the overlapping of biotope types (simultaneous availability of the benefits of more than one habitat element) and/or the greater richness

of vegetation in the edge zone. For the case of roe deer, which typically frequents hedgerows and forest stand edges, our investigations (Gossow, 1986; Reimoser, 1986; Reimoser, 1994b) have shown that edge zones play an important role. However, it appears that it is the separation of biotope types that is important for roe deer, not the overlapping as postulated by Leopold. It was noted that roe deer do not feel at ease in forests with irregular and diffuse, eye-level vegetation, even if such areas offer the most varied and best supply of food. They prefer optically striking edges, which appear to assist their orientation (a function of the physiology of the eyes of roe deer (Reimoser, 1986; Reimoser, 1994b), as well as providing good thermal and hiding cover. If differentiated biotope types bounded by sharp edges are in close proximity, this unnatural, over-attractive habitat provokes deer over-abundance and, hence, game damage.

Applicability: Our study results and the theoretical consequences concerning predisposition to game damage relate to montane and sub-alpine site conditions (in Austria, above 700 m altitude). The consequences for lowland sites may be different (e.g. where regeneration growth is much faster, there is less snow in winter, etc.).

5. Conclusions

The important conclusion is that unacceptable game damage is promoted by poor forestry practices. To counter this, forestry must rehabilitate monocultural forests to be more natural, and natural regeneration strategies must be the preferred ones. Clear cutting with a need for artificial reafforestation should be avoided, thereby further reducing the predisposition of a forest to game damage.

It has been shown that such forest management is economically viable, indeed often more profitable than clear-cut management: however, provision of related information and training for foresters and hunters alike must be intensified. At the same time the forester achieves an optimal forest environment that satisfies the hunting fraternity, the public, and the demands of tourism also.

6. Recommendations

1. See the forest ecosystem, including its animals, holistically.
2. Recognize the importance of the silvicultural system as a habitat factor.
3. Include ungulate game as a site factor in forest management.
4. Recognize the importance of the silvicultural system as a game-damage factor.
5. Define management targets clearly to be able to recognize 'damage'.
6. In assessing damage, take compensatory mortality of trees into account.
7. Reduce food-independent settling stimuli.
8. Improve natural food availability.
9. Avoid clear cuts and optically striking stand edges.
10. Promote natural regeneration in silviculture.
11. Coordinate habitat and game management.
12. Support more interdisciplinary research into forest–ungulate–man interactions.

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